

MUTHIAH SHANMUGAM

SENIOR CAE ENGINEER — VEHICLE DYNAMICS & CONTROLS SIMULATION

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PROFILE

Results-driven Senior CAE Engineer with 11+ years of experience in full-vehicle multibody simulation, ride comfort and handling analysis, and vehicle dynamics model validation for OEM passenger car programs — including control system integration for active chassis systems under SiL/MiL co-simulation environments, enabling virtual validation without physical prototype hardware.

CORE COMPETENCIES & TECHNICAL SKILLS

- ▶ Full-Vehicle MBS Simulation (ADAMS/Car, SIMPACK)
- ▶ SiL/MiL Co-Simulation & Controls Integration (ADAMS/Controls, Simulink)
- ▶ Ride Comfort, Handling & K&C Analysis
- ▶ Driveline & Subsystem MBS (ADAMS/View, Gear AT)
- ▶ Python Simulation Automation & Tool Development

MBS Simulation: ADAMS/Car · ADAMS/View · ADAMS/Controls · SIMPACK · Gear AT

SiL/MiL & Co-Simulation: MATLAB/Simulink · FMU/FMI · Damper control

Programming: Python 3 · PyQt · C/C++ · GitHub · Tortoise SVN

FEM: HyperMesh · ANSA · SFE Concept

PROFESSIONAL EXPERIENCE

Senior CAE Engineer — Vehicle Dynamics & Controls Simulation — *Akkodis (formerly AKKA Technologies / MBtech Group GmbH & Co. KGaA)*

Sindelfingen, Germany | April 2015 – Present | Technical lead on 1–2 concurrent projects, teams of 2–3 engineers

Full-Vehicle MBS — K&C, Ride Comfort and Handling

- ▶ Built and evaluated full-vehicle models in ADAMS/Car and SIMPACK across all development phases — from concept through series maturity.
- ▶ Correlated simulation results against rig and road test data to series-level accuracy; identified deviation root causes and derived model improvements.
- ▶ Actively supported technical alignment with OEM customers and represented results in design reviews.

SiL/MiL Controls Integration — Active Chassis Systems

Model-based development of closed-loop SiL/MiL co-simulation environments for virtual validation of active chassis controllers.

- ▶ Model-based development of MBS co-simulation models (ADAMS/Car, ADAMS/Controls, Simulink) via SiL/MiL for virtual validation of active chassis controllers.
- ▶ Designed and integrated control algorithms including black-box subsystems; defined signal architecture and closed control loops.

Subsystem Modeling, Validation and Controls Integration in MBS

- ▶ Developed and validated ADAMS/View subsystem models for driveshafts, transmission components, park mechanisms and differentials.
- ▶ Integrated Simulink control logic into ADAMS/View for co-simulation and validation of control strategies.
- ▶ Simulated and analyzed noise-critical operating conditions in the differential — drove concept optimization based on results.

Python Simulation Automation & Process Manager

- ▶ Developed a GUI-based simulation process manager in Python (PyQt) to automate K&C, ride and handling simulations in ADAMS/Car.
- ▶ Developed comparison/analysis functions for KPIs; integrated parameter optimization and sensitivity analysis for chassis tuning.

Master's Thesis — MBS Department — *MBtech Group GmbH & Co. KGaA*

Germany | Sep 2014 – Feb 2015 | "Simulation-based development and adaptation of control strategy for semi-active damper on full vehicle model"

- ▶ Developed a control strategy for semi-active dampers on a full-vehicle model via co-simulation between MATLAB/Simulink and ADAMS/Car.

Werkstudent — Digital Product R&D, Mercedes-Benz Cars — *Daimler AG*

Böblingen, Germany | Dec 2013 – Aug 2014

- ▶ Contributed to a digital development environment for vehicle door structures; performed mesh adaptation for rapid design changes.

Project Engineer — Engineering Design Services (FEM) *Wipro Technologies*

Bangalore, India | Jun 2011 – Jul 2013

- ▶ Developed FE models for crash, occupant safety and pedestrian safety analysis (HyperMesh, ANSA).

EDUCATION

M.Eng. Automotive Systems — Vehicle Dynamics | Hochschule Esslingen, Germany | 2013–2015 | Grade: 1.8 — Focus: MBS, ride & handling, control theory

B.Eng. Mechatronics | Anna University of Technology, India | 2008–2011 | CGPA: 9.62/10 — Focus: modeling, simulation, control theory

PUBLICATIONS

Shanmugam, M., Demeester, E., Tjahjowidodo, T. (2025). "Comparative study of fixed and adaptive control strategy in semi-active suspension systems using a quarter car model." AIP Conf. Proc. 3204, 030008. DOI: 10.1063/5.0246102

Shanmugam, M., Demeester, E., Tjahjowidodo, T. (2023). "Semi-active suspension control using P-LQR method." AIP Conf. Proc. 2914, 030003. DOI: 10.1063/5.0175961

Muthiah S., Balasubramanian K., Balusamy V. (2011). "An Investigation to Eliminate the Hot Cracking Tendency in Super Austenitic Stainless Steel." Intl. J. Multidisciplinary Research Academy. ISBN: 13-978-81-920297-0-2

LANGUAGES

English — C2 (Fluent) · German — B2 (Professional working proficiency) · Tamil — Native